

MEC203: Engineering Graphics and CAD

Spring 2026

Autodesk Fusion 360 Project -- 3D Modeling of a Mars Rover

1. The Overall Design Requirements

You are the lead designer for NASA JPL. NASA intends to launch the concept design of its next-generation Mars Rover next month. The overall design will be similar to that used in their present product. You are invited to design new wheels, Mobility Systems, Rover Chassis, Acquisition Arm, and lastly assemble them in a Rover.



Above are pictures of the 3D assembly of the current product. If your new design is more elegant, your manager will be thrilled. The Mars Rover is car-sized, about 10 feet long (not including the arm), 9 feet wide, and 7 feet tall (about 3 meters long, 2.7 meters wide, and 2.2 meters tall). The diameter of the wheels is 50cm.

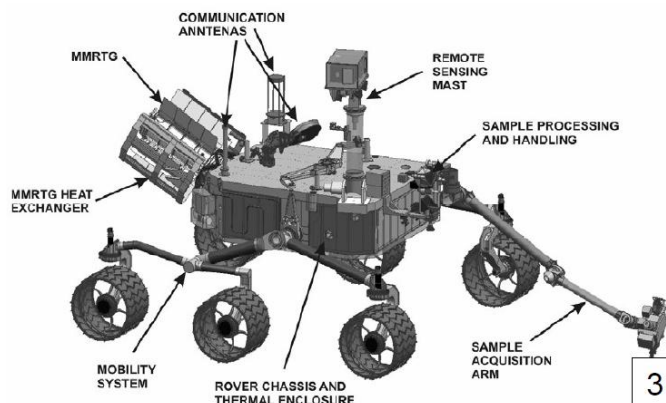


Figure 1. An illustration of Mars Rover Components.

Please consider the following requirements in order of precedence:

- 1) All parts need to be able to function together properly. Interference and undesired gaps in the assembly should be avoided.
- 2) The shape and dimensions of each part need to be practical.
- 3) Style is important but lightweight is essential.
- 4) Function, manufacturability, and economy should be considered while designing.
- 5) You are only responsible for designing the general shape of the parts. For example, you don't need to design a small bearing with detailed cylindrical rollers inside.

If you do have detailed parts, which will of course make your design more attractive (getting higher grades), please show them out as your highlight feature in your introduction video.

- 6) The overall size of the rover should be less than 3x3x3(m).
- 7) **Make motion studies.** Please make at least **3** different kinds of motion studies (e.g., wheel revolving, door sliding/opening, or acquisition arm extension). You should make it like it is running in slow motion (of course the rotation of the wheels should not be too fast to observe).

More motion studies will boost your grade. Without motion studies, you will get almost zero for the final project. Please highlight your motion studies in your video.

- 8) All above will be considered part of your project's grading criteria.

If you need help, please contact our TAs.

2. Select your final project group

Please sign up for your final project group on Brightspace under "Groups". Here are two important things you need to know about selecting a group.

- 1) If you decide to work by yourself, please sign up for the "Final Project by Yourself". This will stop others from joining your group.
- 2) If you want to work with a team, please sign up for the "Final Project by Group". Each group can only have **up to 2 students**. You should have permission if you want to join another student's group. We have created a forum on Brightspace under "Discussion". Feel free to post a thread to find group members. If you try to contact the other group members, the normal email address will be firstname.lastname@stonybrook.edu. **(If you join another's group without any permission and the other members of the group claim that you did nothing, you will be removed from the group and given a zero for the final project.)**
- 3) Please sign up as soon as possible on Brightspace for your project group. **Remember you should have formed your group by Friday, April 17.** If you have any doubts about forming a group, please contact us as soon as possible.
- 4) Note: you will get a **10% bonus point** if you work by yourself (single group). However, based on past experience, please keep in mind that due to the workload of the final project, working alone does not always lead to a higher score. Group work often results in better designs and higher-quality models through shared ideas and collaborative effort. **Choose wisely.**

3. Overall guidelines/checklist for the final project

- 1) Read carefully all files included in the project before you start, to be sure about what you will have to do.
- 2) Design all the components for your Mars Rover.
- 3) Make the assembly of the Mars Rover. Check for interference. **Use correct/necessary constraints and fit and tolerance.**
- 4) **Make motion studies.** You should make it like it is running in slow motion (of course the rotation of the wheels should not be too fast to observe).
- 5) Finish the "Project Report" to summarize your design based on the template (which can be found on Brightspace).
- 6) Make an introduction video. When the design is completed, **prepare an introduction video to show your work.**

4. Detailed requirement

- 1) In the video, you are expected to show how you came up with the idea, how you transfer your idea to a 3D model design, how your design looks like, how it works (the motion studies), and everything else worth saying. **Remember to mention and explain the fits and tolerances that you gave to your design for the connection parts.** Your introduction video should be less than 5 minutes.
- 2) This video needs to be uploaded to YouTube and you need to paste the video link to the project number assignment excel. **The link will be posted later on Brightspace.**

(Federal law of FERPA-Family Educational Rights and Private Act protects the privacy of the students so you **MUST** not include any type of personal information inside the introduction video. **Please show and ONLY show your group number at the beginning of the video.** Other types of personal information, including but not limited to names, SBU IDs, SSNs, etc. **must not appear in the video.**)

If you choose to speak during the video, make sure your voice can be heard clearly. If you choose to type sentences in the video, try not to block the entire screen with a whole paragraph. Keep it concise and straight to the point.

Use any software you can find online to create the introduction video.

- 3) After finishing everything, please zip all the necessary files to one zipped folder and upload it to Brightspace. **Only one of the members of the group needs to submit it.** You can create a zipped folder from Windows (select the folder that includes all the **necessary** files, right click→Send to→Compressed (zipped) folder). Tip: Try to unzip and open it on a different PC to make sure that everything works before submission. If you open the files directly from a zipped folder, it might not work properly.
- 4) Do not forget to paste your link when you make your final submission onto Brightspace.
No late submission will be accepted unless there is a documented emergency. Computer issues, internet issues, or any other technical issues are not considered "documented emergencies".

5) Final Project Grading Breakdown (100 pts total)

Your final project will be evaluated based on the following components:

1. Design and Modeling – 30 pts

- Accuracy and completeness of individual parts
- Proper use of constraints and dimensions
- Clean and logical modeling strategy

2. Motion Study – 30 pts

- Correct assembly and joint definitions
- Realistic and functional motion behavior
- Demonstration of mechanism performance

3. Video Quality – 15 pts

- Clear and well-organized introduction video
- Effective demonstration of design and motion
- Good visibility, narration (if applicable), and presentation

4. Project Report – 25 pts

- Clear explanation of design process and decisions
- Inclusion of screenshots, key steps, and analysis
- Overall organization, clarity, and professionalism

6) **Contact Zhengke, Ran, Yixue via email if you any questions.**

5. Detailed Final Project Timeline & Deadlines

Remember you should have formed your group **by Friday, April 17 On Brightspace.**

Deadline 1: Modeling + Overall Assembly: **11:59 PM, Friday, May 1, 2025.**

Deadline 2: The whole project (Modeling + Overall Assembly + Motion studies + Introduction Video + Report etc.): **11:59 PM, Wednesday, May 13, 2025.**

Upon each deadline, you should upload the required parts/files to Brightspace. Failing to meet **Deadline 1** will result in **20% penalties** in your total final project grade. Failing to meet **Deadline 2** will result in a **0** for your total final project grade.

You can change/modify your designs after your submission on **Deadline 1**. The final grading will be based on the final submission on **Deadline 2**.

Tips:

- Make sure that you upload the correct files on Brightspace on time. **Not-working/no motion studies or badly assembled designs will get almost zero.**
- **Late projects will not be accepted unless there is a documented emergency.**
- **Computer issues, internet issues, or any other technical issues are not considered “documented emergencies”.**

Please pay attention to the deadlines (The deadline for submitting the whole project \ introduction video link).